SELF-DRIVING CAR

by
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CPE161P-4_C1
REAL-TIME EMBEDDED SYSTEM

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INTRODUCTION

Objective Overview:

- **To demonstrate how a self driving car works (Forward/Backward).**
- **To be able apply what you have learned in logic circuits and switching theory, microprocessors, and CPE160P.**
- **To be able to apply real-time embedded system using arduino kit.**

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EXPERIMENT OVERVIEW

- Components Used:
  - Arduino Kit
  - 2-WD Car Chassis Kit
  - IR Sensor
  - IR Remote

- Self-driving Car Feature:
  - Remote-Control Operated
  - Directional Movement Control

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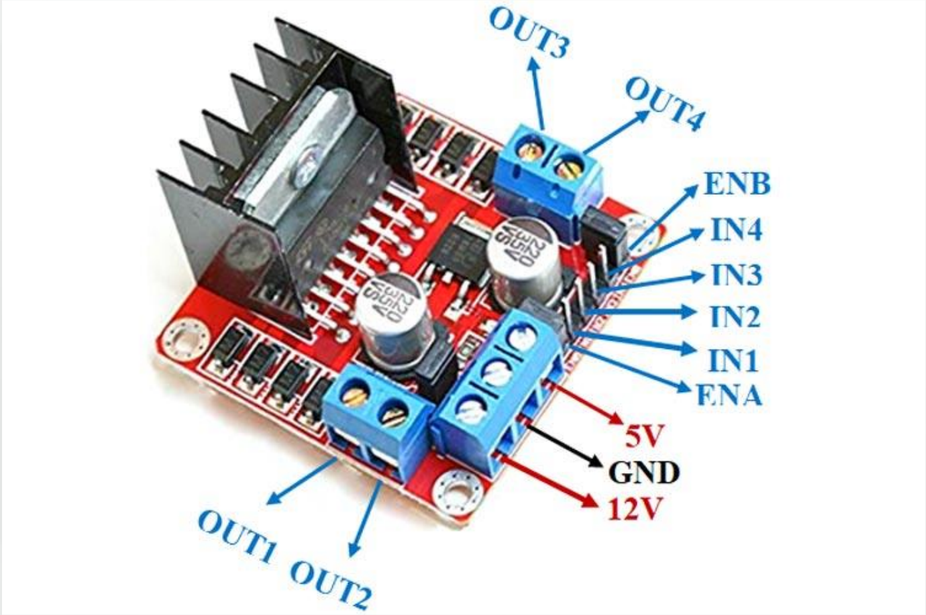
DISCUSSION - MATERIALS PIN OUT



L298 DC MOTOR DRIVER

WHEEL + DC MOTOR

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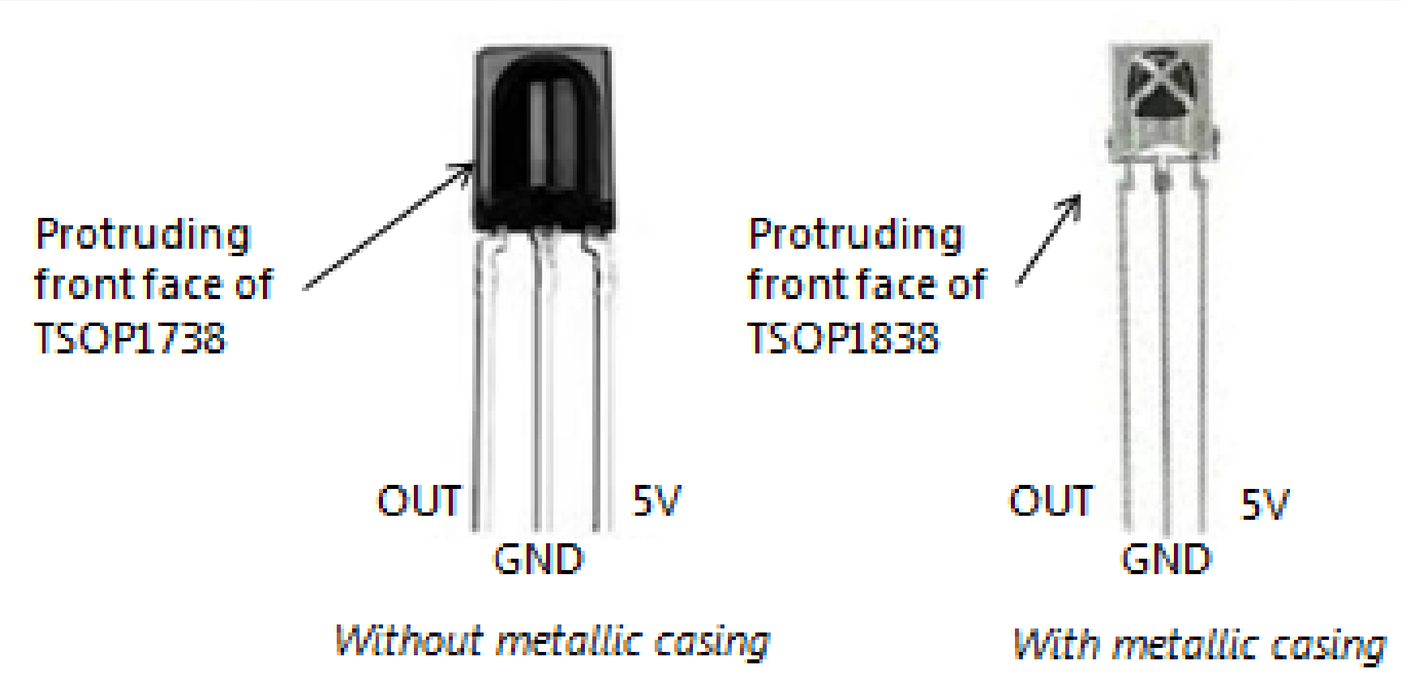
DISCUSSION - MATERIALS PIN OUT



IR RECEIVER

DISCUSSION – ASSEMBLING AND RESEARCHING

- Circuit Construction:
  - Assemble Whole Car set
  - Assembled components: 2 DC Motor, Battery, L298 DC Driver, IR Receiver
  - Wired All component together
- Research Component Integration
  - Research How L298 Driver Works
  - Check existing code for reference

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DISCUSSION – CODING

- Code Construction:
  - Use previous code as guide for ir remote
  - Use existing code for car motor as a reference
  - Integrate both logic together
  - Test the Movement of Car Wheel
  - Debug if error persist

DISCUSSION – REMOTE KEY CODE

```
/* Key   Code
 CH-   BA45FF00
 CH    B946FF00
 CH+   B847FF00
 <<    BB44FF00
 >>    BF40FF00
 >||   BC43FF00
 -     F807FF00
 +     EA15FF00
 EQ    F609FF00
 100+  E619FF00
 200+  F20DFF00
 0     E916FF00
 1     F30CFF00
 2     E718FF00
 3     A15EFF00
 4     F708FF00
 5     E31CFF00
 6     A55AFF00
 7     BD42FF00
 8     AD52FF00
 9     B54AFF00 */
```

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DISCUSSION - INITIALIZATION

```
#include <IRremote.hpp>

//Initialize Car Wheel Motor Pin
int in1 = 3;
int in2 = 4;
int in3 = 5;
int in4 = 6;

//Define Button Code
#define ForwardButton 0xE916FF00 // Code for Forward Button
#define ReverseButton 0xF30CFF00 // Code for Reverse Button
#define LeftWheelButton 0xBB44FF00 // Code for Left Wheel Button
#define RightWheelButton 0xBF40FF00 // Code for Right Wheel Button
#define StartButton 0xBC43FF00 //Start Button
```

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DISCUSSION – Setup

```
void setup() {
  //Remote Initialization
  Serial.begin(9600);
  IrReceiver.begin(8, ENABLE_LED_FEEDBACK);
  Serial.println("Ir Receiver Ready");

  //Pin Mode for Motor
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);

}
```

DISCUSSION - Loop

```
void loop() {
  if (IrReceiver.decode()){
    uint32_t decodedValue = IrReceiver.decodedIRData.decodedRawData;
    Serial.print("Decoded Data: ");
    Serial.println(decodedValue, HEX);

    if (decodedValue == ForwardButton){
      Forward();
      delay(1000);
      digitalWrite(in1, LOW);
      digitalWrite(in2, LOW);
      digitalWrite(in3, LOW);
      digitalWrite(in4, LOW);
    }
    if (decodedValue == ReverseButton){
      Reverse();
      delay(1000);
      digitalWrite(in1, LOW);
      digitalWrite(in2, LOW);
      digitalWrite(in3, LOW);
      digitalWrite(in4, LOW);
    }
```

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DISCUSSION - Loop

```
if (decodedValue == LeftWheelButton){
  LeftWheel();
  delay(1000);
  digitalWrite(in1, LOW);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, LOW);
}
if(decodedValue == RightWheelButton){
  RightWheel();
  delay(1000);
  digitalWrite(in1, LOW);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, LOW);
}
```

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DISCUSSION - Loop

```
  if(decodedValue == StartButton){
    Forward();
    delay(1000);
    digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
    digitalWrite(in3, LOW);
    digitalWrite(in4, LOW);
    delay(200);
    Reverse();
    delay(1000);
    digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
    digitalWrite(in3, LOW);
    digitalWrite(in4, LOW);
    delay(200);
    LeftWheel();
    delay(1000);
    digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
    digitalWrite(in3, LOW);
    digitalWrite(in4, LOW);
    delay(200);
    RightWheel();
    delay(1000);
    digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
    digitalWrite(in3, LOW);
    digitalWrite(in4, LOW);
  }
  IrReceiver.resume();
}
```

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DISCUSSION – Directional Movement Functions

```
void Forward(){
  digitalWrite(in1, LOW);
  digitalWrite(in2, HIGH);
  digitalWrite(in3, LOW);
  digitalWrite(in4, HIGH);
}

void Reverse(){
  digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
  digitalWrite(in3, HIGH);
  digitalWrite(in4, LOW);
}

void LeftWheel(){
  digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
}

void RightWheel(){
  digitalWrite(in3, HIGH);
  digitalWrite(in4, LOW);
}
```

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DISCUSSION - INTEGRATION AND TESTING

- **Test Wheel Direction Functionality:**
  - Verify if Wheel works in the right direction properly.
- **Test IR Remote Functionality:**
  - Verify if Input is being read
  - Check if wheel move in direction pressed in key

CONCLUSION

- **Objectives Achieved:**
  - Demonstrated how a how a self driving car works (Forward/Backward).
  - Applied what i have learned in logic circuits and switching theory, microprocessors, and CPE160P.
  - Apply real-time embedded system using arduino kit.

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